BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 063 · Strings: Concatenation and Slicing

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does assignment do to a variable?

Q2. What is NOT true?
A. true
B. 0.5
C. false
D. An empty string

Q3. What values does a Boolean variable represent?

Q4. Computers share files on one school site. Two sites in different towns also need a link. Correct both claims. Explain why. (a) A LAN spans several countries. (b) A WAN only works in one room.

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I Do · Read new knowledge

Key vocabulary

Concatenation — The + operator joins two strings in order without automatically inserting a space.

Substring — substring(start,length) returns characters beginning at a zero-based index.

Length — The .length property gives the number of characters in a string.

Indexing — The first character in OCR ERL is at position zero.

Syntax — The rules governing valid programming statements.

Test data — An input used to check expected results.

New learning

Concatenation: The + operator joins two strings in order without automatically inserting a space. Apply this in a welcome message.

Substring: substring(start,length) returns characters beginning at a zero-based index. Apply this in a name badge.

Length: The .length property gives the number of characters in a string. Apply this in a club code.

Indexing: The first character in OCR ERL is at position zero. Apply this in a ticket suffix.

Worked example

A welcome message is being developed. Explain how concatenation and substring are used. [4 marks]

1. Read the scenario and identify the data the program needs.

2. Concatenation [1]; The + operator joins two strings in order without automatically inserting a space. [1].

3. Substring [1]; substring(start,length) returns characters beginning at a zero-based index. [1].

4. Two linked points for each concept.

We Do · Practise with guidance

Guided 1

A name badge uses concatenation. Explain how it helps. [2 marks]

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Worked method: Concatenation [1]; The + operator joins two strings in order without automatically inserting a space. [1].

Guided 2

Explain substring in a club code. [2 marks]

Worked method: Substring [1]; substring(start,length) returns characters beginning at a zero-based index. [1].

Guided 3

A ticket suffix uses length and indexing. Explain both. [4 marks]

Worked method: Length [1]; The .length property gives the number of characters in a string. [1]. Indexing [1]; The first character in OCR ERL is at position zero. [1].

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the substring to display the first three characters of "NETWORK".

```
name = "NETWORK"
print(name.substring(0, ...))
```

write

Question 1

State TWO relevant techniques from this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 2

Explain how substring and length help with a school password hint. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 3

A programmer designs a café menu label. Explain why concatenation, length and indexing matter. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

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Exit Ticket · Answer before checking

The + operator joins two strings in order without automatically inserting a space.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It stores a value in the named variable, replacing its previous value if one existed.

Q2. C — false

Q3. True or false.

Q4. (a) A LAN covers a small area, such as a school site. (b) A WAN links networks over a large area, often using outside providers' links. Internet access alone does not decide the type.

I Do / We Do pages · Guided practice

Q1. Concatenation [1]; The + operator joins two strings in order without automatically inserting a space. [1].

Q2. Substring [1]; substring(start,length) returns characters beginning at a zero-based index. [1].

Q3. Length [1]; The .length property gives the number of characters in a string. [1]. Indexing [1]; The first character in OCR ERL is at position zero. [1].

You Do pages · Code First

```
name = "NETWORK"
print(name.substring(0, 3))
```

Expected result: NET

You Do pages · Independent questions

Q1. Concatenation [1]; Length [1].

Q2. Substring [1]; substring(start,length) returns characters beginning at a zero-based index. [1]. Length [1]; The .length property gives the number of characters in a string. [1].

Q3. Concatenation [1]; The + operator joins two strings in order without automatically inserting a space. [1]. Length [1]; The .length property gives the number of characters in a string. [1]. Indexing [1]; The first character in OCR ERL is

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at position zero. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: